

IIIM-SAM: Zero-Shot Texture Anomaly Detection Without External Prompts

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Abstract—Anomaly detection is a crucial aspect in ensuring the reliability of industrial products in smart manufacturing. Despite employing unsupervised anomaly detection methods to mitigate the challenges associated with obtaining defect data, there remains the challenge of adapting anomaly detectors to drift within normal data distributions, especially when there are significant intra-class variations in normal samples. Consequently, zero-shot anomaly detection techniques, which do not require training data and thus avoid interference from training data, have emerged as a new research direction. Meanwhile, the zero-shot segmentation capability demonstrated by the Segment Anything Mode (SAM) visual foundation model has further fueled our research enthusiasm. In this paper, we focus on texture images and propose a no-external prompt SAM-based method for zero-shot texture anomaly detection, called Image Internal Information Mining SAM (IIIM-SAM). We utilize the Image Internal Information Mining Prompter to mine relationships between internal regions of the image, identifying specific background and potential defect points. The two types of prompt points are fed into the SAM model in different stages, imparting the SAM with incorporating category information segmentation capability. Unlike current methods, our zero-shot texture anomaly detection method requires no external prompts or expert knowledge, and also avoids the uncertainty introduced by text prompts necessary for CLIP-based approaches. Our experimental results demonstrate the superiority of our zero-shot texture anomaly detection method compared to other approaches. Specifically, in the texture subset of MVTecAD/KolektorSDD2, our IIIM-SAM achieves 99.2%/93.6% image-level AUROC and 98.6%/92.3% pixel-level AUROC.

Note to Practitioners—The motivation of this paper arises from a practical challenge in anomaly detection in smart flexible manufacturing—customized customer demands lead to significant intra-class variations and even entirely different types of products. Currently, most anomaly detection methods struggle to adapt quickly to this challenge, severely impeding production efficiency. This paper proposes a zero-shot texture anomaly detection method based on SAM, which automatically generates background and potential defect points through an image internal information mining prompter. Compared to existing zero-shot

anomaly detection algorithms based on foundation models, our method's most significant uniqueness is without any expert knowledge and the absence of any external text prompts. Through comparisons with four advanced methods on six instances using two evaluation metrics, the results demonstrate the outstanding anomaly detection capability of our approach.

Index Terms—Smart manufacturing, texture anomaly detection, prompt engineering, foundation model, SAM.

I. INTRODUCTION

INDUSTRIAL anomaly detection aims to identify defects on the surface of industrial products, which holds significant importance in ensuring product quality in industrial production [1], [2]. In contrast to common computer vision tasks, data for industrial anomaly detection is highly imbalanced and acquiring anomaly data is extremely challenging. Even for normal (non-defective) data of the same product type, significant intra-class variations often occur due to changes in consumer demands. Furthermore, with the widespread adoption of intelligent dark factories and the implementation of flexible manufacturing, there is a growing demand for anomaly detection methods that can rapidly adapt to different types of products. Specifically, as shown in Fig. 1, the automatic optical inspection (AOI) system in flexible manufacturing requires inspecting multiple types of industrial products on the same production line without changing the algorithm of AOI.

Therefore, adaptation to different products with training-free has become a significant challenge in the current field of anomaly detection. Specifically, there are three key points: 1) No need for repeated training according to different products. 2) Effective anomaly detection capabilities across multiple products. 3) Inference without the need for external expert knowledge prompts. Clearly, both conventional supervised anomaly detection methods [3], [4] and the prevailing unsupervised learning methods [5], [6], [7], [8], which fit probability distributions using normal samples, cannot meet the requirements of the latest and challenging demands. Naturally, zero-shot anomaly detection has piqued our research interest. The cold-start characteristic of zero-shot anomaly detection means it can be directly applied to surface inspection without the need for retraining or target samples as references.

In recent years, some foundational vision models have demonstrated surprising capabilities in detecting unknown objects and out-of-distribution data. For instance, the visual-language models [9], [10] have demonstrated outstanding performance in zero-shot classification tasks. Through

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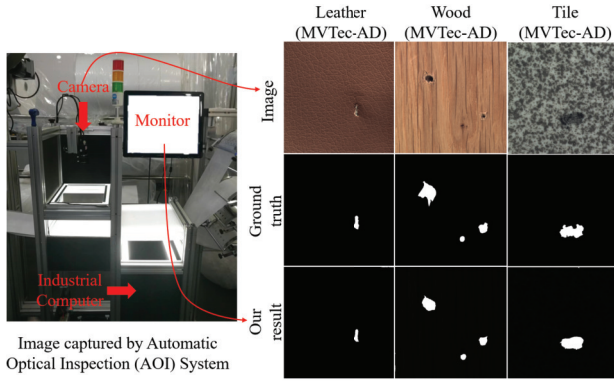


Fig. 1. Zero-shot anomaly detection samples from automatic optical inspection (AOI) system deploying Image Internal Information Mining SAM (IIIM-SAM).

large-scale learning from millions of image-text pairs, they learned expressive representations capturing a wide range of concepts. Such models exhibit remarkable zero-shot transfer capabilities for downstream tasks, including image classification [11], object detection [12], and segmentation [13]. However, it should be noted that these models require appropriate textual prompts including inspected categories, inspected status, and more detailed description templates to extract knowledge effectively, ensuring optimal performance in downstream tasks. Numerous studies have indicated that the performance of visual-language models varies significantly depending on the different text prompts. However, using appropriate text prompts to inject external expert knowledge into the model increases the skill requirements for operators in practical applications. It also significantly limits the methods' ability to quickly adapt to the surface inspection across different products. Therefore, the problem we aim to address is large-model-based zero-shot anomaly detection which is free from the constraints of external prompts.

Segment Anything Model (SAM) is another open-source foundational vision model, which exhibits exceptional generalization capability to segment any object in any image without additional training [14]. This advancement provides a new research foundation for zero-shot anomaly detection. Nevertheless, when SAM is directly applied to anomaly detection, it exhibits two significant gaps: 1) SAM is an interactive framework that requires predefined prompts such as points, bounding boxes, or masks. This does not align with the zero-shot anomaly detection requirement in real-world scenarios, where no external expert knowledge is available. 2) It is a category-agnostic segmentation method, which prevents us from clearly discerning whether the segmented result represents a defect or background local structure. Specifically, we expect the SAM to perceive and segment the entire image as defect-free after receiving background prompt points. However, as illustrated in Fig. 2 (a), even when we provide accurate background prompt points (indicating no defect information), the native SAM still segments out local regions in the defect-free image, resulting in false positives. These limitations make SAM unsuitable for fully automatic comprehension of industrial product surface images. Additionally, we observe that

background texture interference with distinct edges in anomaly detection and the lack of well-defined defect edges pose significant challenges to SAM's segmentation capabilities, as shown in Fig. 2 (b). Overall, the performance of SAM in anomaly detection largely depends on the type, location, and number of prompts.

In this paper, we focus on texture anomaly detection [15], [16] and strive to achieve the first fully automated zero-shot anomaly detection without external prompts, as shown in Fig. 3. To leverage SAM's sensitive target perception ability in anomaly detection and make it practically valuable, it is necessary to investigate how to obtain accurate prompts without external knowledge. Secondly, we design a suitable anomaly detection framework based on SAM.

The motivation for our internal image information mining prompt method is based on two observations regarding defect features compared to defect-free features of the same image: 1) The inter-class relationship difference is greater than the intra-class relationship difference. 2) Defect features are fewer than defect-free features. We propose a prompter network for internal information mining, which combines the advantages of convolutional structures and transformer structures. By dividing the input image into blocks, the network explores the mutual relationships between blocks and judges defect-free regions and potential anomaly regions based on the principles of approximating the maximum and the strongest approximate relationship. Thus, two types of prompts are provided. We further design a two-stage mask decoder architecture based on the difference in confidence levels between the two types of prompts to achieve a more effective anomaly detection framework. As an example, Figure 1 illustrates our approach qualitative detection results for a few cases.

To summarize, our main contributions are:

- 1) *Zero-Shot Anomaly Detection without External Prompts:* To the best of our knowledge, our proposed method is the first large-model-based zero-shot anomaly detection that does not require external textual prompts. Unlike existing methods, our approach eliminates the reliance on external prompts, does not require additional training, and does not need defect samples as references. Therefore, our algorithm is more universal and generalizable.
- 2) *Prompt Points Mining Principle:* We propose a principle for discriminating regions based on the relationships between internal regions of the image. By jointly considering the strength of inter-region relationships and the number of strong relationships, the principle determine whether the area is the defect-free background or the defective foreground. This principle allows for the mining of prompt points for downstream large models based solely on the inspected input image.
- 3) *Prompter Network Design:* We design an internal information mining prompter network (IIIM-prompter) based on prompt points mining principle. IIIM-prompter organically combines the advantages of convolutional structures and transformer structures, and utilizes the weight information of the attention mechanism to represent the interrelationships between regions. Through the aforementioned design, the IIIM-prompter can more

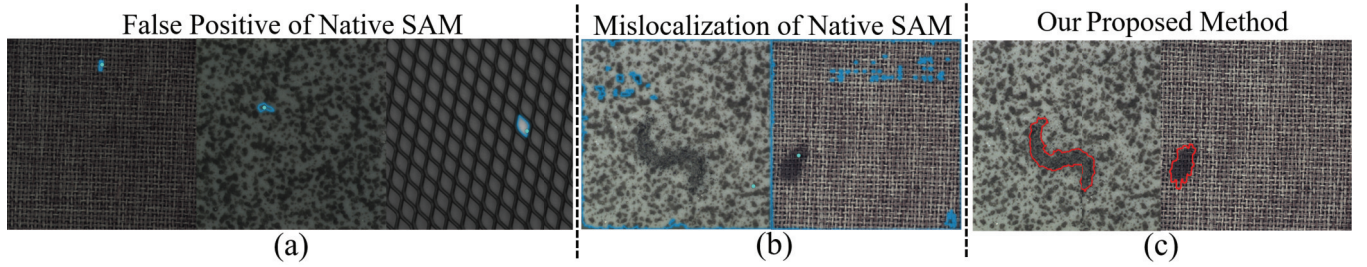


Fig. 2. The results of anomaly detection on the native SAM with manually provided point prompts for defect-free or anomalous regions, where the blue boundaries represent the segmented regions. (a) For defect-free images, providing prompts with defect-free information can also achieve segmentation results (False Positive Examples); (b) For defect images, the interference from background texture can severely impact the anomaly localization performance of the native SAM (Mislocalization); (c) Our method can accurately locate defects without manual intervention.

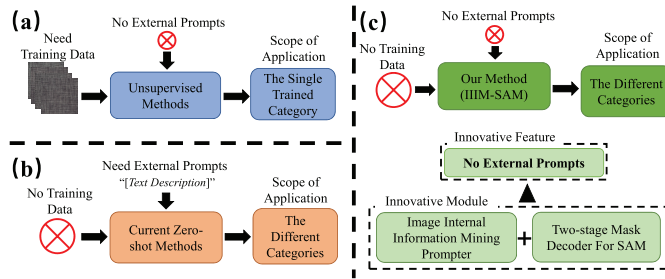


Fig. 3. Characteristics comparison with different anomaly detection methods. (a) shows unsupervised anomaly detection method; (b) shows current zero-shot anomaly detection method; (c) shows the features and unique characteristics of our zero-shot method (IIM-SAM).

accurately provide the SAM model with accurate prompts containing category information without relying on external knowledge or additional dataset training, enabling it to be applied to specific texture defects.

- 4) *Optimization of SAM Architecture for Anomaly Detection:* We develop a two-stage mask decoder architecture that processes the prompts in different stages, effectively utilizing the category information embedded within the prompts. This architecture significantly improves the anomaly detection amidst complex backgrounds, through bidirectional cross-validation of two different types of information (anomaly and background). Experimental results indicate that our approach demonstrates the state-of-the-art (SOTA) performance.

II. RELATED WORK

A. Foundation Model

In recent years, the emergence of foundation models has attracted the attention of almost all artificial intelligence researchers. These models are trained on large amounts of data and exhibit impressive “emergence capabilities” across various scenarios, i.e. zero-shot generalization capabilities [10], [17], such as chat-GPT [18], GPT-4 [19], CLIP [11].

Inspired by the tremendous success of foundation models in the field of Natural Language Processing (NLP), researchers have started building large-scale foundation models for computer vision and exploring the boundaries of zero-shot capabilities of these models. The SAM model [14] uniquely leverages an extensive collection of masks

(11 million images, including over a billion masks) and demonstrates robust generalization capabilities. However, it is a task-agnostic segmentation model that requires interactive prompts, such as prior points, bounding boxes, or masks. This characteristic makes it less conducive to end-to-end automatic target segmentation. In this paper, we explore the potential of harnessing SAM’s general segmentation capabilities for detecting anomalous regions. Additionally, the proposed Image Internal Information Mining Prompter can be extended to other foundational vision models for other downstream tasks.

B. Zero-Shot Anomaly Detection Method

Zero-shot anomaly detection methods often rely on the strong generalization capabilities of models to detect unseen anomalies [16], [20]. Early research in zero-shot anomaly detection was conducted based on CLIP, such as CLIP-AD [21], ZOC [22]. WinCLIP [23] introduces a groundbreaking work, which uses a large number of manually crafted text prompts and involved forward passes for multiple image blocks. Building on WinCLIP, the first-place method of VAND competition [24] introduces a learnable linear projection technique to enhance the modeling of local visual semantics. MVFA [25] is a multi-level feature adaptation and comparison framework, which leverages a pre-trained vision-language model to achieve generalizability in zero/few-shot medical anomaly detection scenarios. However, these methods are influenced by the lack of universally embedded text prompts, reducing their performance in identifying anomalies related to various unseen object semantics. Additionally, the use of text prompts injects extra knowledge into the model. Anomaly-Clip [26] is the most advanced zero-shot anomaly detection method, which significantly improved the issue of information leakage caused by manually preset text prompts by introducing learnable text prompts. LTAD [27] is method for long-tail anomaly detection, which also improves model performance by learning learnable text embeddings. However, these methods still require the presetting of hyperparameters such as the length of the learnable text prompts and necessitates training the learnable part of the text prompts on relatively similar domain data. MuSc [28] utilizes mutual scoring of unlabeled images and designs an innovative zero-shot anomaly detection method. However, the drawback of MuSc is that the inspected images must have a substantial

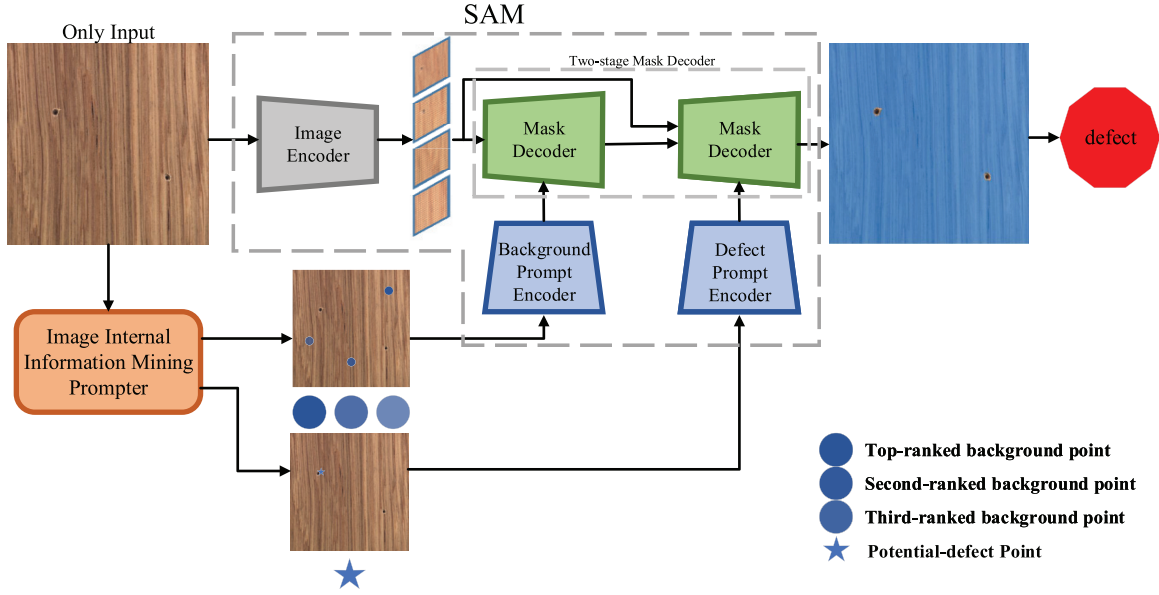


Fig. 4. Overview of the Image Internal Information Mining SAM (IIIM-SAM) structure.

quantity and small intra-class variation to enable effective mutual comparison. AnomalyGPT [29] is a parallel work utilizing foundation model for anomaly detection, but it is designed for unsupervised/few-shot anomaly detection with manually crafted prompts. SAA+ [30] is a zero-shot anomaly detection method based on the SAM foundation model, but unfortunately, it inevitably requires the use of highly specialized external knowledge. In summary, current zero-shot anomaly detection methods all necessitate the injection of external information.

III. BACKGROUND AND CORE PRINCIPLE

A. Zero-Shot Anomaly Detection Setting

In this paper, we want to emphasize a unique zero-shot anomaly detection task setting, where the products are inspected without introducing any external expert knowledge. External expert knowledge includes rich predefined text prompt templates, anomaly appearance description, and so on. This task setting aligns more closely with the automation and rapid production transition requirements in practical applications.

B. Discrimination of Internal Regions Based on Feature Relationships

We recognize that anomalies on the surface of industrial products often result from accidents during the manufacturing process or occasional events due to process instability. Therefore, the number of pixels in anomalous regions is relatively small compared to the anomaly-free regions (i.e., defect-free areas). Even though there may be variations in the surface texture of the product, for a given detection image, the dissimilarity between anomalous and non-anomalous regions must be greater than the dissimilarity between non-anomalous regions from feature statistics perspective. By treating anomalies as foreground and defect-free regions as background,

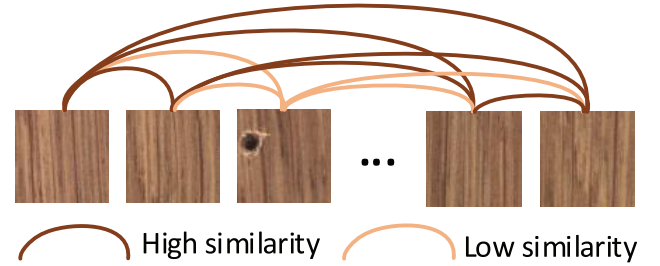


Fig. 5. Principle of internal information mining based on feature relationships.

the anomaly detection problem can be viewed as an image segmentation problem to distinguish between foreground and background. Thus, as shown in Fig. 5 we partition the inspected image, calculate the similarity between regions, and rank them based on the degree and quantity of similarity. This approach allows us to identify multiple definite background regions and one most likely foreground region.

IV. IMAGE INTERNAL INFORMATION MINING SAM

A. Framework Overview

As shown in Fig. 4, Image Internal Information Mining SAM (IIIM-SAM) consists of four main parts: the image encoder, prompt encoder, mask decoder, and image internal information mining prompter. The image encoder, prompt encoder, and each mask decoder are components of the open-source SAM model. The image is processed by the image internal information mining prompter, which provides three definite background prompt points and one potential foreground prompt point based on the strength of region relationships. Considering the varying reliability of information carried by background and foreground prompt points, we design a two-stage mask decoder scheme to validate the anomaly detection results. Finally, based on the segmentation

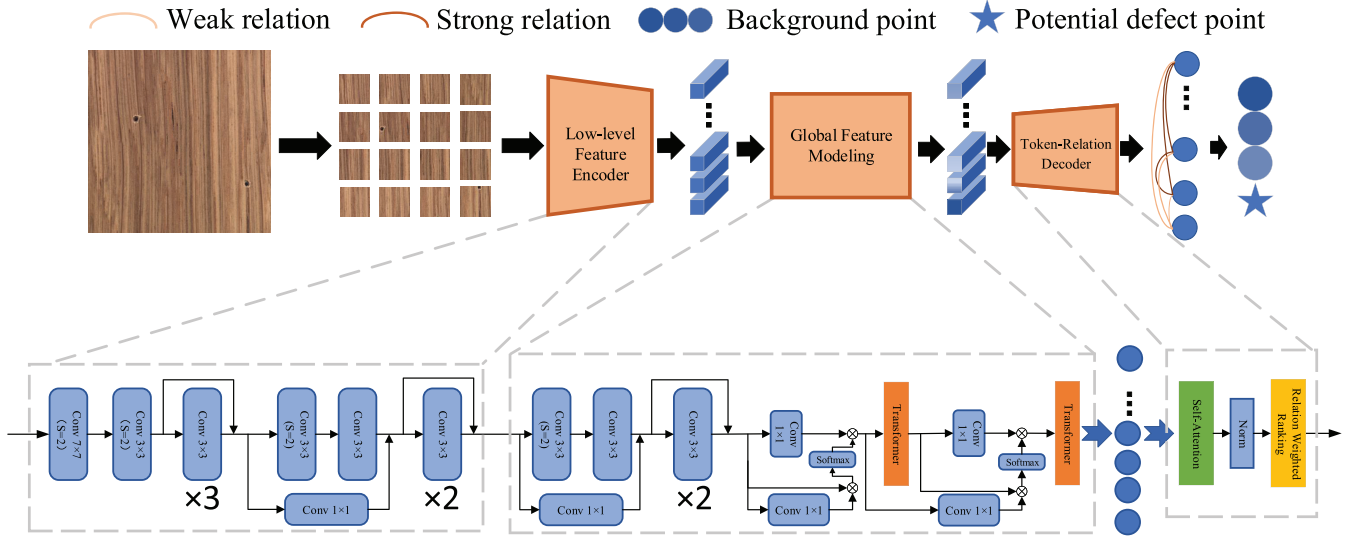


Fig. 6. The detail structure of the image internal information mining prompter. The lower-left part illustrates the internal structure of low-level feature encoder; the middle-bottom part shows the internal structure of global feature modeling, and the lower-right part depicts the internal structure of token-relation decoder.

results of the background and foreground, the decision is made on whether the test image contains defects. The whole process can be expressed as follows:

$$F_{img} = f_{i-encoder}(I), \quad (1)$$

$$P_b, P_f = f_{IIIM-prompter}(I), \quad (2)$$

$$F_{prompter} = f_{p-encoder}(P_b, P_f), \quad (3)$$

$$Result = Det(f_{m-decoder}(F_{img}, F_{prompter})), \quad (4)$$

where I denotes the detected image, F_{img} represents the image features encoded by SAM image encoder, P_b, P_f are sets of prompt points mined by the IIIM-prompter. P_b represents the determined background points, while P_f represents the potential foreground points. Furthermore, $F_{prompter}$ refers to the representation extracted by the prompter encoder, and Det refers to determining the presence of defects based on the segmentation information obtained from the mask decoder.

B. Image Internal Information Mining Prompter

In this subsection, we will provide a detailed introduction to the Image Internal Information Mining Prompter (IIIM-prompter) constructed based on the principles of internal region relationships discussed in Section III. In summary, we first process the image by dividing it into patches and represent the mutual relationships between patches through attention weights in the network. Therefore, we choose the Transformer architecture as the foundation of our prompter network. However, we also recognize that the texture information contained in the defect carrier (non-defective background) often plays a crucial role in discriminating anomaly features. Neglecting the extraction of texture information would be a fatal flaw for anomaly detection. The ability to extract low-level feature (detailed texture information) is evidently a challenge for the Transformer architecture. To address this, we design a prompter network that combines the Convolutional Neural Networks (CNN) with Transformer. We first fully utilize the

effectiveness of convolutional operations in extracting low-level features by performing feature encoding on the input image to form the initial tokens. As shown in the upper part of Fig. 6, the prompter consists of three main parts: Low-level Feature Encoder, Global Feature Modeling, and Token-Relation Decoder.

1) *Low-Level Feature Encoder*: In this stage, we focus on the feature extraction of background texture information. Numerous studies have demonstrated the inherent advantage of the CNN architecture in local information extraction compared to the Transformer architecture [31], [32], [33]. Additionally, the translation invariance of convolutional networks contributes to improving generalization. Therefore, we exclusively employ the CNN architecture in Low-level Feature Encoder to utilize the local information extraction capability and translation invariance characteristics of CNNs. To enhance the representation of low-level features in the encoder's output, we adopt a design with residual skip connections. The specific structure of the low-level feature encoder is illustrated in the lower-left part of Fig. 6.

2) *Global Feature Modeling*: As shown in the detailed structure in the lower part of Fig. 6, for the embedding of low-level features, we further enhance the embedding channel dimensions to increase the dimensions available for evaluating features and facilitate the comparison of token relationships in the next stage. Subsequently, the feature embedding is processed into feature tokens through spatial attention, complemented by two layers of Transformer layers for advanced semantic modeling of low-level features. It is worth noting that, to maintain the stability of spatial positional information, the generation of subsequent spatial attention weights is based on the tokens before the first Transformer layer. The process of Global Feature Modeling can be expressed as follows:

$$F = Conv(f), \quad (5)$$

$$T_1 = SOFTMAX_{HW}(F \cdot A_s(F))^T \cdot F, \quad (6)$$

$$T_T = Transformer(T_1), \quad (7)$$

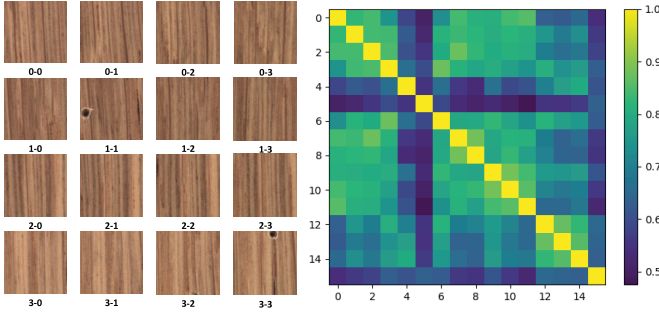


Fig. 7. Visualization of normalized relationships between tokens. On the left are segmented blocks, and on the right are pairwise relationships between blocks converted into tokens. Specifically, each value represents the relationship between the token in the corresponding row and the token in the corresponding column.

$$T_2 = \text{SOFTMAX}_{HW}(T_T \cdot A_s(T_1))^T \cdot T_T, \quad (8)$$

$$T_{out} = \text{Transformer}(T_2), \quad (9)$$

where f are the features extracted from Low-level Feature Encoder, A_s denotes the spatial attention weights. It is important to note that the sources of information for obtaining the spatial attention weights in the two stages are different. The first stage is sourced from F , while the second stage is sourced from T_1 .

3) *Token-Relation Decoder*: After the patch of each image region undergoes processing in the first two stages, it is encoded into tokens corresponding to the number of blocks. These tokens are then collectively fed into the Token-Relation Decoder to distinguish whether each patch belongs to the background or foreground of the image. Specifically, we input the tokens into a standard self-attention layer to extract attention weights A_{ij} between tokens. After normalization, the final relation decoding is accomplished by sorting the relation weights.

In Fig. 7, we present the visualization of normalized relationships between tokens. It is evident that there is a clear distinction in the strength of relationships between regions with and without defects, and strong mutual relationships exist between regions without defects. This further validates the correctness of the principle of distinguishing between defective and background (defect-free) regions, demonstrating the effectiveness of the image internal information mining prompter designed based on this principle.

As formed as Equation 10, we assign equal importance to the strength of relationships and the quantity of strong relationships in the sorting process, aligning with the two distinguishing principles stated in Section III.

$$\text{Relation} = \text{sort} \left(\left(\frac{\sum_j^n A_{ij} \cdot w_{ij}}{\sum_j^n w_{ij}} \right)_{i=1,2,\dots,n} \right), \quad (10)$$

where n is the number of the tokens. The parameter w_{ij} is assigned different weight values based on the strength of the relationship between tokens. Through the variation of this parameter, tokens with higher similarity in relationships are given greater attention.

In the end, after calculating and sorting the relationships between tokens, the regions corresponding to the top three

tokens are considered as three determined background regions. The token at the bottom of the ranking is considered as the potential foreground (defect) region.

4) *IIIM-Prompter Training*: We have designed a contrastive learning training method that is more in line with the goals of internal information mining. Unlike the common practice of applying data augmentation combinations to images within a batch for contrastive learning, we aim to align the information scale during the learning process with the task objectives. Therefore, we propose a training approach, called internal region contrastive learning. We divide an image into N patches, p , and apply different data augmentation methods to these patches to obtain p' . Specifically, we selected the color distortion, random rotation, and Gaussian noise as the data augmentation methods in the training. We then input these pairs (p, p') into the previously mentioned IIIM-prompter. Finally, we obtain a $2N \times 2N$ relationship matrix. By selecting the lower-left $N \times N$ matrix, we acquire the pairwise similarities between p and p' . Each matrix's diagonal position represents the similarity between p and p' , while the others represent the similarity between p and $N-1$ negative examples.

For each row, we apply softmax classification, utilizing cross-entropy loss as the loss function. Ultimately, the loss for internal region contrastive learning is expressed as:

$$\mathcal{L}_p = -\log \frac{\exp(\text{sim}(p, p')/\tau)}{\sum_{i=1}^N \exp(\text{sim}(p, p_i)/\tau)}, \quad (11)$$

where τ is the temperature in the loss.

C. Two-Stages Mask Decoder for Zero-Shot Anomaly Detection

In this section, we will outline the observed segmentation drawbacks of applying the original SAM model to anomaly detection, as well as the design of a two-stage mask decoder structure tailored to address the identified characteristics of the original SAM.

1) *The Segmentation Performance Characteristics of the Original Sam*: Firstly, we observed that SAM tends to struggle in capturing the global background when faced with textures with pronounced edges. It often gets trapped in local structural textures, a phenomenon we refer to as “over-segmentation”, as shown in Fig. 2. Therefore, when dealing with textures exhibiting clear and structured edges, a joint prompting of multiple points is required.

Secondly, while joint prompting with multiple points effectively alleviates the “over-segmentation” issue in SAM, our experiments revealed that certain types of defects without disrupting the texture structure and only color changes, are easily overlooked by SAM. Hence, we design a step that we refer to as “retrospective checking” to mitigate instances of false negatives.

2) *Structural Detail*: Due to the mentioned drawbacks exhibited by the native SAM in texture anomaly detection, we design a two-stage mask decoder structure, as shown in Fig. 4. This structure cleverly combines the uncertainty of background points and foreground prompting in its design. Specifically, since the category of the three background points are determined, the first-stage mask decoder only takes these

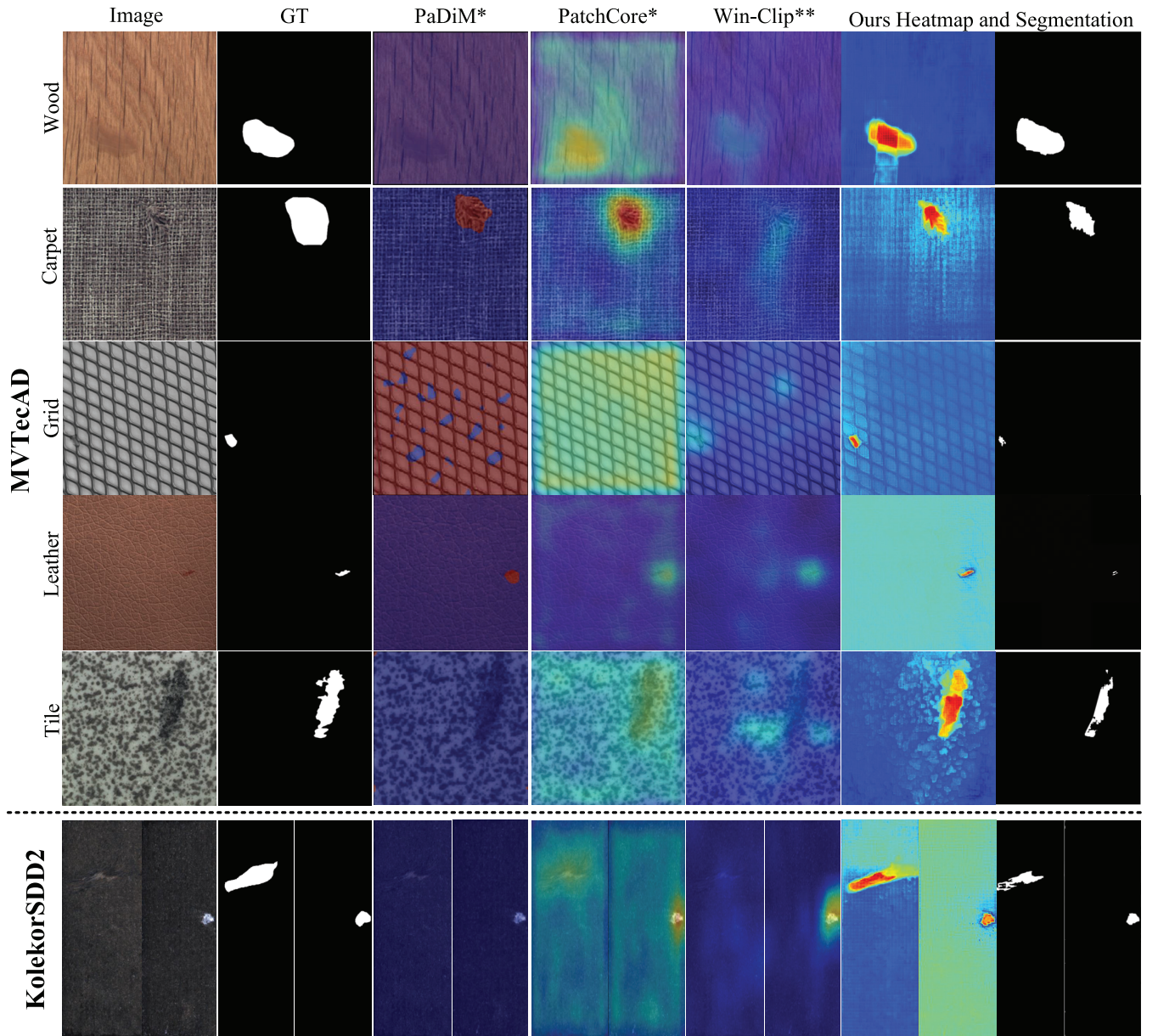


Fig. 8. Qualitative comparison of anomaly detection results on different datasets. * indicates training with only one sample, and ** indicates zero-shot method using simple text prompts, such as, “defective.” Our method is zero-shot without any external prompt.

three background point prompts as input. Subsequently, the second-stage mask decoder, based on the results from the first stage, introduces the potential foreground point prompt for a defect-checking phase. The purpose of retrospective checking is to further confirm the detection results from the first stage of the model, avoiding missed detection of defect region in the first-stage decoded mask. If the defect exists in the image, the lowest-ranked area based on the interrelationship scores within the image will definitely be a defect area. The missed defect will be naturally detected. In cases where the potential foreground point does not belong to the foreground (defect) area, the retrospective check will not be affected either. Because, the decoding process will return large and scattered segmentation results when the proposed point is from background region, which indicates that the image is

indeed defect-free. Ultimately, the judgment of the existence of anomalies is based on the segmentation results verified by the second-stage mask decoder.

D. Anomaly Decision and Segmentation

We summarize the anomaly decision and segmentation pipeline of the proposed method with Pytorch-style pseudocode in Algorithm 1. For images without anomalies, our IIIM-SAM will not segment out local regions and produce the segmentation results without targets (the segmentation map will be entirely black). For images with anomalies, our method will segment out local regions after inputting prompts from IIIM-prompter, and output the segmentation maps with targets.

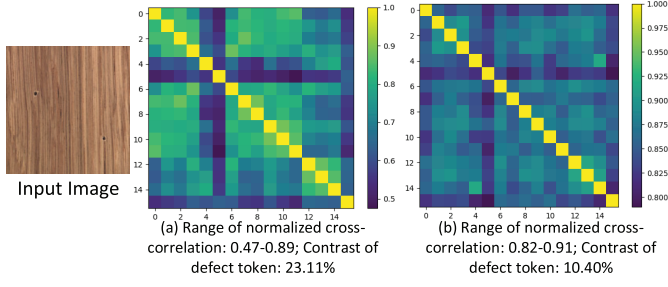


Fig. 9. The comparison of removing low-level feature encoder and global feature modeling. (a) is retaining the two components. (b) is removing the two components and token features are extracted from the SAM image encoder.

Algorithm 1 Pseudo Code for IIIM-SAM

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Input: Image  $\mathcal{I}$ ;
Functions:  $\text{img\_encoder}()$ ,  $\text{bg\_prompt\_encoder}()$ ,
 $\text{defect\_prompt\_encoder}$ ,  $\text{mask\_decoders}()$ ;
Outputs:  $\text{Seg\_Result}$ ,  $\text{Anomaly\_Decision}$ ;
 $F_{\text{img}} = \text{img\_encoder}(\mathcal{I})$ ;
 $\text{Points}_{\text{bg}}, \text{Point}_{\text{defect}} = \text{prompter}(\mathcal{I})$ ;
 $P_{\text{bg}} = \text{bg\_prompt\_encoder}(\text{Points}_{\text{bg}}, \text{Point}_{\text{defect}})$ ;
 $P_{\text{defect}} = \text{defect\_prompt\_encoder}(\text{Point}_{\text{defect}})$ ;
 $\text{Seg\_Result} = \text{mask\_decoders}(P_{\text{bg}}, P_{\text{defect}}, F_{\text{img}})$ ;
if  $\text{Seg\_Result} > 0$  then
  |  $\text{Anomaly\_Decision} \leftarrow$  "Defects present";
else
  |  $\text{Anomaly\_Decision} \leftarrow$  "No Defects";
end

def  $\text{prompter}(\mathcal{I})$ :
  |  $F_{\text{low-level}} = \text{feature\_encoder}(\mathcal{I})$ ;
  |  $F_{\text{global}} = \text{global\_feature\_modeling}(F_{\text{low-level}})$ ;
  |  $\text{Point}_{\text{bg}}, \text{Point}_{\text{defect}} = \text{relation\_decoder}(F_{\text{global}})$ ;

```

V. EXPERIMENTS

We conduct a series of comparative experiments to assess the zero-shot anomaly detection performance of IIIM-SAM. In addition, we compare the zero-shot IIIM-SAM with few-shot anomaly detection methods that require additional training and the state-of-the-art (SOTA) zero-shot anomaly detection methods. We also perform extensive ablation studies to validate the individual effectiveness of the proposed components. The detailed setups, *e.g.*, datasets, metrics, introduction to the compared methods, and other implementation details, are given in this section.

A. Dataset Description and Evaluation Indicators

1) *Dataset Description:* We evaluate our approach in various real-world texture defect detection scenarios, including MVTecAD [34], the whole KolektorSDD2 [35] and DAGM [36] datasets. Both datasets contain high-resolution images. The input resolution is 1024×1024 for MVTecAD and roughly 230×640 for KolektorSDD2 dataset. To ensure a fair comparison, we maintain the same input resolution for both our proposed method and the comparison methods in all experiments.

a) *MVTecAD Dataset:* MVTecAD dataset is the most widely used and authoritative benchmark for anomaly detection. The dataset is divided into two major anomaly detection categories: texture and object. In this study, we focus on texture anomaly detection, consisting of sub-datasets for five product classes, namely Grid, Leather, Tile, Carpet, and Wood. The defects collected in the texture set include various sizes, shapes, and categories, amounting to a total of 25 defect types. Therefore, evaluating our IIIM-SAM and other methods on this dataset provides a comprehensive assessment of algorithm performance.

b) *KolektorSDD2 Dataset:* KolektorSDD2 defect dataset is constructed from images of defected production items that were provided and annotated by Kolektor Group. KolektorSDD2 dataset consists numerous types of defects and their images including 356 images with defects and 2979 images without defect. The images in this dataset are captured by a visual inspection system in a controlled environment, and they have similar dimensions. Defects vary in shape, size, and color, ranging from small scratches and spots to large surface defects. Experiments on KolektorSDD2 dataset demonstrate that texture anomaly detection algorithms are not limited to product types similar to those showcased in MVTecAD. Users can capture surface images of industrial structural components through a professional visual inspection system, enabling the application of our IIIM-SAM method.

c) *DAGM Dataset:* The DAGM dataset is a classic texture defect detection dataset widely used in the field of anomaly detection. It includes a variety of material surfaces and defect types. The defect regions are annotated using segmentation masks in the dataset. Each image in the dataset has a resolution of 512×512 . The characteristics of the DAGM dataset differ significantly from the two previously mentioned datasets. Performance evaluation on DAGM allows for a more comprehensive demonstration of our algorithm's generalization capability.

2) *Evaluation Indicators:* We consider that image-level classification metrics are the primary aspect for evaluating the method, because it is more crucial from a practical perspective. After all, practitioners only need the detection system to accurately identify images containing anomalies for automated defective product removal. So, we use the area under the receiver operating characteristic curve (AUROC) at the image level as the evaluation metric following the literature [37], [38]. To observe potential data imbalance more clearly, we also employ the F1-score at the optimal threshold (F1-max) [39]. To more clearly and comprehensively demonstrate the superiority of our IIIM-SAM, we also use pixel-level AUROC and PRO metrics to evaluate its anomaly localization and segmentation capabilities.

B. Introduction to the Compared Methods

We selected six of the latest and representative anomaly detection methods for comparison with our proposed IIIM-SAM, including PaDim [37] (2021), PatchCore [38] (2022), ACR [20] (2024), WinClip [23] (2023), WinClip+ [23] (2023), and Anomaly-Clip [26] (2024). Overall, PaDim, PatchCore,

and ACR are non-language-guided anomaly detection methods, while WinCLIP and AnomalyCLIP are language-guided anomaly detection methods based on large vision-language-models. Specifically, PaDim and PatchCore are advanced few-shot anomaly detection methods learning from defect-free samples, but they lack zero-shot detection capabilities. ACR is the current state-of-the-art non-vision-language-model-based method for zero-shot anomaly detection. WinClip is the first zero-shot anomaly detection method based on large vision-language-model (VLM), marking a milestone in the field. WinCLIP+ is an improved version of WinCLIP. This method enhances detection capability by utilizing few-shot normal samples, enabling more efficient defect localization. Anomaly-CLIP is the latest state-of-the-art VLM-based zero-shot anomaly detection method. Therefore, the five selected comparison methods provide a comprehensive performance evaluation from different perspectives against our proposed method. The comparison with these five methods fully demonstrates that our IIIM-SAM not only performs zero-shot detection without the need for external expert knowledge, such as text prompts, but also achieves optimal performance in zero-shot detection.

It is worth noting that, to compare more fairly with our zero-shot method without external information prompts, we only allow PaDim and PatchCore methods to be trained with one and three defect-free images, respectively (i.e., 1-shot and 3-shot). For the zero-shot anomaly detection method WinClip, we restrict the method to use rich information text prompts and only allow WinClip to use simple and generic prompt texts, such as “defect” or “anomaly”. The above settings apply to all our experiments.

C. Implementation Details

We implement our models in Python 3.10 and PyTorch 1.13. Experiments are run on 2 Nvidia RTX 4090 GPU on Ubuntu 18.04 Linux. For all experiments, we choose the sam-vit-h weights as our pre-trained SAM. IIIM-prompter is trained on the ImageNet dataset [40] and trained by stochastic gradient descent (SGD) optimizer with Nesterov momentum. We use an initial learning rate of 0.01, a momentum of 0.9, and a weight decay of 0.0001. We train for 150 epochs with the learning rate multiplied by 0.1 at 90 and 120 epochs. For the attention values greater than 0.5 in the Token-Relation Decoder, we set w_{ij} to 5, 4, 3, 2, and 1, respectively.

D. The Qualitative Anomaly Detection Results Comparison on Different Datasets

We first present visualizations of anomaly detection using four state-of-the-art methods on different datasets. As shown in Fig. 8, we display detection cases for six product categories on two datasets. The first five rows show cases from the MVTecAD dataset, while the last row presents two cases from the KolektorSDD2 dataset.

Qualitatively, it is evident that our zero-shot IIIM-SAM method exhibits excellent generalization and anomaly detection capabilities. According to some failure cases in the four

TABLE I
AREA UNDER ROC IN % FOR ANOMALY DETECTION OF MVTecAD DATASET ON IMAGE-LEVEL. † INDICATES THE ACCURACY REPORTED IN THE PAPER

Image AUROC	PaDim		PatchCore		easy promp.		ACR†	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Carpet	98.9	99.3	98.6	99.2	81.9	81.5	99.5	100	99.7
Grid	50.9	67.5	52.3	67.7	49.9	48.6	97.9	97.0	98.2
Tile	87.8	93.2	98.4	98.5	76.1	75.8	97.6	100	98.8
Leather	98.6	98.9	97.3	97.9	98.6	98.2	99.1	99.8	99.4
Wood	96.8	97.2	98.3	98.6	92.7	92.2	97.2	96.8	99.8
Avg.	86.6	91.2	89.0	92.4	79.8	79.3	98.3	98.7	99.2

TABLE II
F1-MAX IN % FOR ANOMALY DETECTION OF MVTecAD DATASET ON IMAGE-LEVEL

F1-max	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Carpet	95.1	96.6	94.9	95.3	89.4	89.2	98.8	99.5	99.3
Grid	84.5	85.3	86.2	87.3	86.1	85.7	90.9	96.1	97.2
Tile	90.7	91.5	97.5	97.6	85.9	85.1	97.1	99.5	97.6
Leather	96.2	97.7	95.9	97.5	96.5	96.3	96.4	98.9	98.7
Wood	94.3	94.9	96.1	96.6	91.9	91.6	96.2	95.7	99.4
Avg.	92.2	93.2	94.1	94.9	90.0	89.6	95.9	97.9	98.4

comparative methods (Padim in Wood, Grid, Tile, KolektorSDD2; Patchcore in Grid; WinClip in KolektorSDD2), we demonstrate a more sensitive anomaly detection capability.

Specifically, it can be observed that the PaDim and PatchCore methods experienced varying degrees of failure in anomaly localization on the wood, grid, tile, and KolektorSDD2 datasets. This is because these methods require a large amount of anomaly-free image samples for learning and do not have inherent zero-shot detection capabilities. The WinClip method, using a generic text prompt, showed some anomaly response capability in the presented examples, with only one case of anomaly response failure. However, it is evident that the anomaly heatmaps of WinClip is weak and diffused, making it difficult to distinguish from the background response. This demonstrates that zero-shot anomaly detection methods based on vision-language models, like WinClip, must rely on injecting a large amount of matching expert knowledge through text prompts to work effectively. In contrast, our IIIM-SAM method eliminates the need for external prompts and leverages the strong perception capabilities of the improved SAM model. As a result, our method is more versatile. We can accurately locate defects across different products, and produces highly distinguishable anomaly response peaks (with a clear distinction from background response).

E. The Quantitative Anomaly Detection Results

1) *Results on MVTecAD:* We first presented the quantitative anomaly detection results on the MVTecAD dataset. Table I and Table II show the anomaly detection performance measured using image-level AUROC and F1-MAX metrics, respectively. We compared our method with other zero-shot and one-normal-shot anomaly detection methods. The first five rows represent the image-level metrics for five texture

TABLE III

AREA UNDER ROC IN % FOR ANOMALY DETECTION OF MVTECAD DATASET ON PIXEL-LEVEL. † INDICATES THE ACCURACY REPORTED IN THE PAPER

Pixel AUROC	PaDim		PatchCore		easy promp.		ACR†	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Carpet	97.8	98.3	98.4	98.5	82.7	77.9	97.8	98.8	98.5
Grid	70.2	73.9	58.8	68.4	75.2	61.1	90.4	97.3	98.1
Tile	84.1	88.3	94.1	94.6	79.8	65.0	90.3	94.6	97.9
Leather	98.5	98.8	98.8	99.0	96.9	94.3	98.6	98.6	99.0
Wood	90.7	92.2	92.7	93.3	91.6	89.2	89.2	96.5	99.5
Avg.	88.3	90.3	88.6	90.8	85.2	77.5	93.3	97.2	98.6

TABLE IV

PRO IN % FOR ANOMALY DETECTION OF MVTECAD DATASET ON PIXEL-LEVEL

PRO	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Carpet	92.9	94.2	93.3	93.8	81.2	71.1	88.4	90.1	95.0
Grid	41.2	50.1	21.7	30.4	79.8	52.0	81.3	75.6	92.9
Tile	66.7	72.8	82.5	83.0	89.4	51.2	75.7	87.6	91.3
Leather	95.6	96.3	95.2	96.2	92.9	91.1	88.9	92.2	94.1
Wood	86.5	88.2	87.1	87.7	90.7	70.1	80.5	91.2	97.7
Avg.	76.6	80.3	76.0	78.2	86.8	67.1	83.0	87.3	94.2

subcategories, while the last row shows the average detection results across all categories.

It can be observed that our IIIM-SAM method performs exceptionally well, achieving superior average performance under both evaluation metrics. Specifically, for the image-level AUROC metric, our method achieves detection scores above 98% in all five categories, while other methods show significant performance disadvantages in at least one category. In terms of average performance, our method holds a clear advantage, outperforming the WinClip method, which uses simple prompts, by 19.9%. Even compared to Anomaly-CLIP, which requires an auxiliary training set to optimize text prompts, we achieve a 0.5% improvement. The balanced performance of our method is even more evident under the F1-MAX metric, which effectively avoids the evaluation bias caused by unbalanced test data. For example, other comparison methods show a significant performance drop in the wood category. This further comprehensively validates the superior anomaly detection performance of the IIIM-SAM method.

Next, we used pixel-level AUROC and PRO metrics to measure the anomaly region segmentation ability of our method and the comparison methods. The anomaly segmentation results of Table III and Table IV demonstrate the advantage of our IIIM-SAM method for anomaly localization. Specifically, we surpassed four categories in pixel-level AUROC, with the highest lead of 3.3% compared to the next best method, and an overall average lead of 1.4%. In terms of the pixel-level PRO metric, we achieved comprehensive superiority. This reflects that once our method detects the presence of an anomaly, it locates the anomaly region with high precision. In contrast, even when other comparison methods correctly detect anomalies, they exhibit more erroneous responses in the anomaly regions. In other words, compared to other state-

TABLE V

AREA UNDER ROC IN % FOR ANOMALY DETECTION OF KOLEKTORSDD2 DATASET ON IMAGE-LEVEL

Image AUROC	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Kolektor SDD2	58.5	63.6	91.4	92.5	77.4	75.0	88.2	91.0	93.6

TABLE VI

F1-MAX IN % FOR ANOMALY DETECTION OF KOLEKTORSDD2 DATASET ON IMAGE-LEVEL

F1-max	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Kolektor SDD2	22.7	27.1	73.9	77.0	44.1	43.1	82.6	88.3	90.5

TABLE VII

AREA UNDER ROC IN % FOR ANOMALY DETECTION OF KOLEKTORSDD2 DATASET ON PIXEL-LEVEL

Pixel AUROC	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Kolektor SDD2	62.7	66.5	90.4	91.7	77.9	68.2	89.9	78.9	92.3

TABLE VIII

PRO IN % FOR ANOMALY DETECTION OF KOLEKTORSDD2 DATASET ON PIXEL-LEVEL

PRO	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
Kolektor SDD2	51.6	60.0	66.7	69.2	76.0	60.9	81.1	72.6	86.7

of-the-art methods, our approach offers higher interpretability and reliability in determining defective samples.

2) *Results on KolektorSDD2*: The KolektorSDD2 dataset includes more challenging samples for anomaly detection. The defects in the KolektorSDD2 dataset are more complex and less visually distinct from the background compared to other datasets. This complexity makes it difficult for various methods to accurately identify and detect defects. From Table V and Table VI, we can observe severe performance degradation in terms of AUROC and F1-max for the five compared methods. Specifically, language-guided VLM-based zero-shot anomaly detection methods struggle to adapt and effectively detect anomalies because the morphology and semantics of defects are more abstract and diverse, making it difficult for text descriptions to provide effective prompts. However, since our IIIM-SAM does not rely on external prompts and includes a self-derived prompting method, our approach still demonstrates competitive detection performance, consistently maintaining evaluation results above 90%.

As shown in the Table VII and VIII, the advantage of mining prompt information within the image is further reflected in pixel-level metrics, which indicate anomaly localization and segmentation capabilities. Compared to the next best zero-

TABLE IX

AREA UNDER ROC IN % FOR ANOMALY DETECTION OF DAGM DATASET ON IMAGE-LEVEL

Image AUROC	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
DAGM	66.7	70.4	74.2	79.7	83.7	81.8	97.1	97.5	98.5

TABLE X

F1-MAX IN % FOR ANOMALY DETECTION OF DAGM DATASET ON IMAGE-LEVEL

F1-max	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
DAGM	75.9	80.2	80.6	82.5	76.0	75.5	94.8	95.3	96.9

TABLE XI

AREA UNDER ROC IN % FOR ANOMALY DETECTION OF DAGM DATASET ON PIXEL-LEVEL

Pixel AUROC	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
DAGM	80.0	85.5	86.5	89.9	88.1	78.6	91.9	95.6	97.3

TABLE XII

PRO IN % FOR ANOMALY DETECTION OF DAGM DATASET ON PIXEL-LEVEL

PRO	PaDim		PatchCore		easy promp.		ACR	Anomaly -Clip	Our
	1-shot	3-shot	1-shot	3-shot	WinClip+ (1-shot)	WinClip			
DAGM	74.6	77.4	77.8	79.2	81.5	62.7	83.3	91.0	92.9

shot method, IIIM-SAM outperforms with a 2.4% lead in pixel-level AUROC and a 5.6% lead in PRO metric. The experimental results on the challenging KolektorSDD2 dataset further demonstrate the stability and generalization of our method, showcasing its stronger zero-shot detection capabilities.

3) *Results on DAGM*: To further demonstrate that our method's anomaly detection capability is independent of product categories, we also conduct experiments on the DAGM dataset. As shown in Table IX–XII, our method is superior to other SOTA methods in terms of both anomaly detection and anomaly location capabilities. Specifically, compared with the second-best method, our method achieves at least 1% improvement on the anomaly classification (image-level) metric and more than 1.7% improvement on the anomaly localization (pixel-level) metric. These results further highlight the robustness and excellent generalization ability of our approach for texture anomaly detection, attributed to the image internal information mining mechanism without relying on external information learning.

F. Ablation Studies

1) *The Impact of Removing Low-Level Feature Encoder and Global Feature Modeling*: We excluded the Low-level Feature Encoder and Global Feature Modeling from IIIM-prompter, and instead directly used features extracted by the SAM image

TABLE XIII

THE IMPACT OF DIFFERENT NETWORK STRUCTURE CONFIGURATIONS ON MINING INTER-REGIONAL RELATIONSHIPS OF IIIM-PROMPTER

Configurations of IIIM-prompter		Contrast (Anomaly / Background)
Low-level Feature	Global Feature	
CNN	Transformer	0.278
CNN	CNN	0.139
Transformer	Transformer	0.055

encoder. Taking an example with Wood images containing hole defects, Fig. 9 illustrates the visualization of token relationships for IIIM-prompter with retained Low-level Feature Encoder and Global Feature Modeling (part a) and IIIM-prompter with SAM's image encoder as a replacement (part b). It is evident that the IIIM-prompter with retained Low-level Feature Encoder and Global Feature Modeling (part a) shows a more pronounced difference between defective tokens and non-defective tokens. Specifically, the contrast degree of defect token is 23.11% and the range of normalized cross-correlation is from 0.47 to 0.89. Conversely, using SAM's image encoder to replace Low-level Feature Encoder and Global Feature Modeling in IIIM-prompter (part b) exhibits a less clear distinction in token relationships. Specifically, the contrast degree of defect token is just 10.40% and the range of normalized cross-correlation is from 0.82 to 0.91. Through this ablation study, it is shown that our IIIM-prompter combining Low-level Feature Encoder and Global Feature Modeling module has a stronger ability to mine the interrelationships within image regions.

2) *The Impact of Different Network Structure Configurations on Mining Inter-Regional Relationships of IIIM-Prompter*: To further demonstrate the advantage of the combined convolutional and Transformer structure on mining internal image relationships, we conducted the ablation experiments on the IIIM-prompter using the MVTecAD dataset. We compared the combined structure with networks using either CNN or Transformer alone. The difference in regional relationships was quantitatively represented by the contrast of inter-class (anomaly and background) similarity to intra-class (background and background) similarity. For the experimental results, a higher value indicates a more pronounced difference. The average results of the experiments are shown in Table XIII. Compared with the configurations of CNN or Transformer only, the CNN union Transformer configuration yields the highest value. Specifically, the configuration using only CNN lacks the ability to model global relationships, resulting in a 0.139 decrease in the similarity ratio. On the other hand, using only the Transformer structure clearly lacks the ability to extract local low-level features and spatial invariance, leading to a similarity ratio lower than 0.1. This indicates that there is almost no distinction between anomalies and the background. By combining the strengths of both the CNN and Transformer structures, our method can achieve a clearer resolution of the anomaly regions.

3) *The Impact of Background Prompt Point Number*: We investigated the impact of different numbers of background

TABLE XIV

THE IMPACT OF BACKGROUND PROMPT POINT NUMBER ON ANOMALY DETECTION RESULTS FOR DIFFERENT DATASETS

Number of Background Prompt Points	MVTecAD		KolektorSDD2	
	AUROC	F1-max	AUROC	F1-max
3	99.1	98.4	93.6	90.5
2	95.1	91.9	93.6	90.6
1	70.6	68.8	92.9	90.4

TABLE XV

THE IMPACT OF REMOVING THE POTENTIAL FOREGROUND PROMPT POINT ON ANOMALY DETECTION RESULTS FOR DIFFERENT DATASETS

Potential Foreground Prompt Point	MVTecAD		KolektorSDD2	
	AUROC	F1-max	AUROC	F1-max
✓	99.1	98.4	93.6	90.5
×	96.7	94.2	89.4	85.3

prompt points on the final detection performance. Table XIV presents quantitative anomaly detection results for IIIM-SAM on the MVTecAD and KolektorSDD2 datasets using three, two, and one background prompt points. The results indicate that, on the MVTecAD dataset, using two background prompt points leads to a certain decrease in detection performance, while using only one background prompt point results in a more noticeable performance decline. The reason for this performance drop is that SAM's segmentation performance tends to be limited to local structure interference of the complex backgrounds, and increasing the number of prompt points encourages SAM to have a global perception of the tested image. Since the KolektorSDD2 dataset's texture background lacks interference from local structures, changing the number of background prompt points does not significantly impact the detection results.

4) *The Impact of Removing the Potential Foreground Prompt Point:* Table XV shows the impact of removing potential foreground prompt points on the anomaly detection capability of IIIM-SAM. On the MVTecAD and KolektorSDD2 datasets, both AUROC and F1-max indicators exhibit a significant decrease in detection performance. The input of potential foreground prompt points corresponds to the second stage of the two-stage mask decoder in the IIIM-SAM structure. The input of potential foreground prompt points is designed to avoid SAM ignoring anomaly regions with complex backgrounds due to multiple background prompt points and serves as a verification. Therefore, removing potential foreground prompt points in IIIM-SAM makes it susceptible to missing defects, such as color similar to the background or light color on product surface, leading to an overall decrease in detection performance.

G. Performance Evaluation in Real-World Industrial Scenarios

We evaluated the performance of our method on a real industrial nonwoven fabric inspection line, as shown in Fig. 1. The factory often uses nonwoven fabric materials with different textured backgrounds for mask production based on

customer requirements. Ensuring that the nonwoven fabric is free of defects is crucial for guaranteeing the medical protective performance of the produced masks. During a trial run of the inspection line, a total of 1611 images were analyzed, and our method achieved an accuracy of 98.94%, a missed detection rate of 1.93%, and a false alarm rate of 0.84%. The primary cause of missed detections was that certain oil stains became indistinguishable from the background after drying. In general, the detection performance of our method has met the performance requirements of practical application in nonwoven fabric inspection.

VI. CONCLUSION

In this paper, we introduced a zero-shot texture anomaly detection method using image internal information mining prompt. The key uniqueness of this method is without relying on any external expert knowledge to zero-shot anomaly detection, which effectively addresses the challenges of rapid product changes and iterations in industrial production. The convenience of our IIIM-SAM method is primarily reflected in its usage after deployment, eliminating the need for external prompts or expert intervention, thus making it more convenient and scalable for industrial applications in the real world. We began by presenting the principles of image internal information mining, followed by the design of the Image Internal Information Mining Prompter to operationalize these principles. In addition, we discussed the limitations of the native SAM model when applied to texture anomaly detection and overcame these challenges by using multiple background and foreground points for joint prompting through a two-stage mask decoder. The experimental results indicate that our proposed IIIM-SAM outperforms current CLIP-based zero-shot anomaly detection methods. Additionally, compared to SAM with manually input defect and background location prompts, there is only a localization performance difference of no more than 0.5% in terms of pixel-level AUROC. In the test of inference speed, our method achieves 2.2 FPS on the computer with single NVIDIA RTX 4090 GPU. In future deployment, depending on the actual production requirement, the faster SAM variants can be used to accelerate the computation speed.

In fact, we believe that large foundation models already possess sufficient perceptual capabilities for the environment. The reason they may not be directly applicable to specific tasks lies in the lack of organized and inferred perceptual information for those tasks. Through the IIIM-SAM architecture, we aim to showcase the true zero-shot anomaly detection capabilities of large foundation models. Of course, our IIIM-SAM also has its limitations. Our method can only be effectively applied to texture anomaly detection scenarios, such as fabric manufacturing, wooden flooring, and steel rail inspection. The primary limitation stems from the fact that our method relies on internal relationships within the image to mine prompt points. However, in object anomaly detection scenarios (e.g., defects in products with complex structural variations, such as toothbrushes, cables or others with mechanical parts), the method faces challenges. This is because objects often have diverse local structures and internal components, which can

weaken the internal relationships between regions. Consequently, our method struggles to identify reliable background and foreground prompt points, making it difficult to guide SAM to accurately segment anomalous regions. The experiments on the object categories of the MVTecAD dataset also show that our method's average image-level AUROC is only 30.4%, which is 63.1% lower than the current SOTA method. In future work, we will further explore potential strategies to extend the applicability for zero-shot anomaly detection without introducing external knowledge. For example, introducing more image preprocessing stages to transform object anomaly detection into texture anomaly detection, and introducing more feature information to accurately mine the prompt points in complex object structures.

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